

1 Surface mass balance inference based on a transient glacier 2 evolution model inversion.

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6 **ABSTRACT.** The distributed surface mass balance (SMB) of mountain glaciers
7 is the key forcing for glacier evolution and for projections of meltwater avail-
8 ability. Yet direct observations remain confined to sparse stake networks, and
9 geodetic estimates rest on the assumption of a stationary ice flux. Here, we
10 present a new SMB inference method, implemented within the open-source
11 Instructed Glacier Model (IGM), that recasts the problem as a fully differen-
12 tiable transient inversion. The scheme uses observed surface velocities to ini-
13 tialise a dynamically consistent glacier model, then inverts multi-temporal sur-
14 face elevation changes to estimate the spatial distribution of mass balance that
15 best explains the observed glacier evolution. We apply the method to three
16 well-studied glaciers in the Alps—Argentièrre, Great Aletsch and Rhône—
17 evaluating all three with a single, glacier-independent pipeline. Validating
18 the inferred SMB against in situ stake measurements shows that the surface-
19 velocity misfit tracks the SMB misfit. These results validate our method and
20 render it transferable to any glacier with accurate multi-temporal DEMs and
21 surface velocities, even in the absence of stake measurements, opening a path
22 towards regional-scale, distributed SMB inference.

METHODS

Overview

The observational inputs to our method are two surface digital elevation models (DEMs) of the glacier, acquired at epochs t_1 and t_2 several years apart, together with a surface-velocity field observed near t_1 . From these we infer the glacier’s surface mass balance (*smb*) with a two-step pipeline (Fig. 1), implemented within the open-source Instructed Glacier Model (IGM; ?). The elevation change between the two DEMs is the signal used to recover the *smb* (Sect.). At its core is the depth-integrated mass-conservation equation, which links this geometry change to the *smb*,

$$\frac{\partial H}{\partial t} = \text{smb} - \nabla \cdot \mathbf{q}, \quad \mathbf{q} = H \bar{\mathbf{u}}, \quad (1)$$

where H is the ice thickness, *smb* is the surface mass balance, q is the flux and $\bar{\mathbf{u}}$ is the depth-averaged horizontal velocity. The velocity is supplied by IGM’s deep-learning ice-flow emulator (??), trained on a large ensemble of higher-order flow solutions to return $\bar{\mathbf{u}}$ from the ice geometry and the flow parameters. IGM can evaluate the emulator in two modes—online, in which its weights are updated during the simulation, and offline, in which they are frozen; we use the offline mode throughout, and refer the reader to ? and ? for details.

Equation (1) makes the central challenge explicit: the geometry evolution depends jointly on the unknown *smb* and on the ice dynamics through the flux $\mathbf{q}$. Inferring the *smb* from an observed geometry change therefore requires the dynamical state to be fixed beforehand; otherwise an error in the dynamics would be aliased into the inferred *smb*. An inconsistent initial state triggers an initialisation shock (?) that our short (up to ten-year) windows cannot absorb, unlike long spin-up simulations. Our pipeline reflects this dependence directly (Fig. 1): (i) we first calibrate the dynamical state by data assimilation against a surface-velocity snapshot near t_1 , inverting the ice-thickness field H (Sect.). (ii) Holding the calibrated state fixed, we then infer the *smb*: starting from the t_1 DEM, we integrate (1) forward to t_2 and seek the *smb* whose simulated geometry best matches the observed one at t_2 (Sect.).

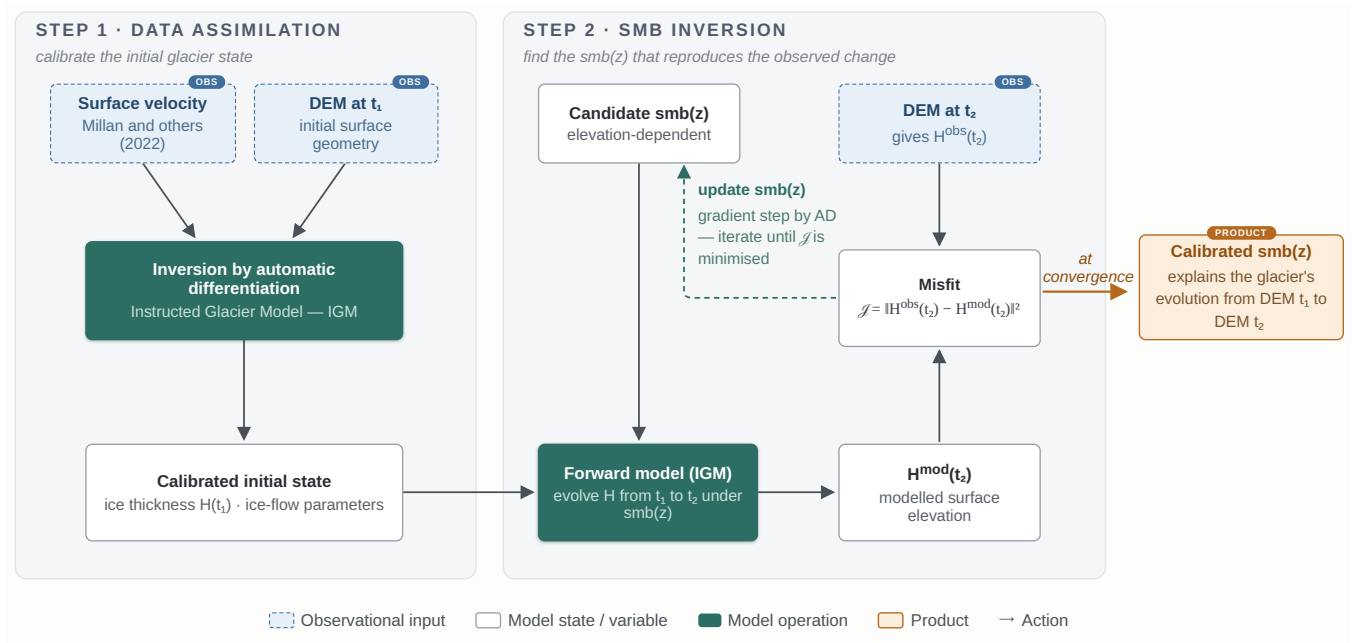


Fig. 1. Overview of the two-step SMB-inference workflow. *Step 1 (data assimilation)*: the observed surface velocity and the start-epoch DEM are assimilated to recover the ice thickness H_{t_1} —and hence the bedrock—while the scalar flow parameters (τ_{ref}, A) are selected by an ensemble search that minimises the surface-velocity misfit. *Step 2 (SMB inference)*: with the calibrated state held fixed, the transient model is integrated from the start DEM to the end DEM under a candidate elevation-dependent profile $smb(z)$; reverse-mode automatic differentiation through the whole forward run yields the profile whose modelled end-of-window geometry best reproduces the observed one. The ice-flow emulator (frozen weights) supplies the depth-averaged velocity at every step.

Table 1. Input and validation data for the three study glaciers. DEM epochs bracket the inversion window and velocities drive the data assimilation. The stake column gives the number of continuous-record locations averaged for the sRMSE.

	Argentière	Great Aletsch	Rhône
Window	2012–2021	2009–2017	2007–2016
Surface DEMs	Pléiades	swisstopo	swisstopo
Velocity		(?)	
Stake locations	25	2	5

Data

We evaluate the method on three Alpine glaciers that span a wide range of size and are among the most intensively surveyed in the world (Table 1): Great Aletsch (Swiss Alps), the largest glacier of the Alps ($\approx 80 \text{ km}^2$); Rhône (Swiss Alps), a mid-sized valley glacier ($\approx 15 \text{ km}^2$) that feeds the eponymous river; and Argentière (French Alps, Mont-Blanc massif), a smaller, steep and fast-flowing valley glacier ($\approx 12 \text{ km}^2$). Each combines repeat DEMs, satellite surface-velocity fields and a decades-long in situ stake network—maintained by swisstopo and GLAMOS in Switzerland and by GLACIOCLIM in France—which supply both the fields that drive the inversion and the ground truth that validates it.

For each glacier the pipeline requires two surface DEMs bracketing the inversion window—whose difference is the geometry change to be explained—and a surface-velocity field for the data assimilation (Sect.). Ice-thickness soundings are used only where available, as an optional refinement (Appendix ??). For Argentière the two DEMs are the 2012 and 2021 Pléiades models of ?; for Great Aletsch (2009, 2017) and Rhône (2007, 2016) they are swisstopo geodetic DEMs. We resample all fields onto a common 100 m grid. Surface velocities are the satellite feature-tracking mosaics of ?, built from Sentinel-2 and Landsat-8 image pairs over 2017–2018. Because alpine surface velocities vary slowly relative to the decadal DEM windows (?), this single field serves as a representative snapshot for the thickness calibration.

For validation we use in situ stake balances, kept strictly independent of the inversion. Rather than compare against every raw stake reading—noisy and irregularly sampled in space and time—we retain only locations with a continuous multi-year record, average each into a single balance value ($\pm$ interannual standard deviation), and evaluate the inferred smb against these averages. This yields two such locations on Aletsch and five on Rhône; on Argentière the denser GLACIOCLIM network provides the published 2012–2021 mean balance at each of 25 stakes.

Ice-dynamics calibration via data assimilation

The first step of the pipeline uses two of the observational fields introduced above—the surface velocity and the t_1 DEM—to recover the ice-thickness field that places the glacier in a dynamically consistent state, free of the initialisation shock described in Sect. .

We use IGM’s existing data-assimilation module (?) to invert for the thickness H by minimising the regularised misfit

$$\mathcal{J}_{\text{DA}}(H) = \alpha_u \|\mathbf{u}_s(H) - \mathbf{u}_s^{\text{obs}}\|^2 + \mathcal{R}(H), \quad (2)$$

where $\mathbf{u}_s(H)$ is the modelled surface velocity returned by the emulator (?), $\mathbf{u}_s^{\text{obs}}$ is the observed surface velocity (?), and $\mathcal{R}$ is a Tikhonov smoothness regulariser on H . The smoothness weight is set at the corner of the resulting L-curve (?). The minimisation is solved by L-BFGS (?); we refer the reader to ? for the full formulation, weighting strategy and convergence behaviour.

The emulated flow regime is not set by the geometry alone: it also depends on two scalar physical parameters, the basal-motion parameter τ_{ref} and the Arrhenius factor A . Here τ_{ref} is the reference basal shear stress in Weertman’s sliding law (?)—the shear stress sustaining a reference basal speed of 100 m yr^{-1} , in MPa—so that a smaller τ_{ref} corresponds to easier sliding. Rather than invert these pointwise, we fix them beforehand by an ensemble search: candidate (τ_{ref}, A) pairs are drawn log-uniformly from $[0.05, 2.0] \text{ MPa}$ and $[30, 200] \text{ MPa}^{-3} \text{ yr}^{-1}$ by a Tree-structured Parzen Estimator (TPE; ?), as implemented in Optuna (?), and each pair is passed through the data assimilation; we retain the pair that minimises the surface-velocity misfit, so the assimilation itself inverts the thickness H alone. We apply this identical procedure to all three glaciers, which keeps the pipeline free of glacier-specific tuning.

Data assimilation inverts the thickness at the start epoch t_1 , which we denote H_{t_1} . Finding the thickness fixes the bedrock,

$$z_b(\mathbf{x}) = z_s^{\text{obs}}(\mathbf{x}, t_1) - H_{t_1}(\mathbf{x}), \quad (3)$$

which provides the substrate on which the transient run subsequently evolves. Only once the calibrated state matches the observed velocity to within observational uncertainty do we accept it and proceed to the SMB inversion.

SMB inference by automatic differentiation

Observation target. With the dynamical state calibrated (Sect.), the smb inversion is driven by the geometry change between the two DEM epochs. The bedrock is already fixed through Eq. (3); combining it with the observed end-of-window surface DEM gives the target thickness

$$H^{\text{obs}}(\mathbf{x}, t_2) = z_s^{\text{obs}}(\mathbf{x}, t_2) - z_b(\mathbf{x}). \quad (4)$$

The two surface DEMs $z_s^{\text{obs}}(\cdot, t_1)$ and $z_s^{\text{obs}}(\cdot, t_2)$ are the glacier-specific two-epoch elevation models of Sect. ; their difference defines the geometry change the inversion must reproduce.

Forward model. With the calibrated thickness H_{t_1} and the flow scalars (τ_{ref}, A) held fixed, the glacier is evolved from t_1 to t_2 by integrating (1) with an explicit forward-Euler scheme,

$$H^{n+1} = \max(0, H^n + \Delta t^n [smb^n - \nabla \cdot \mathbf{q}^n]), \quad (5)$$

where the non-negativity projection prevents spurious negative thicknesses near the margin. The integration starts from $H^0 = H_{t_1}$ and advances until the accumulated time reaches t_2 ; the number of steps N_t is therefore not prescribed but follows from the step sizes selected along the way. Each step is limited by a Courant–Friedrichs–Lewy condition,

$$\Delta t^n = \min\left(c_{\text{eff}} \Delta x / \max_{\Omega} \|\bar{\mathbf{u}}^n\|, \Delta t_{\text{max}}\right), \quad (6)$$

with $c_{\text{eff}} = 0.2 < 1$ ensuring stability and $\Delta t_{\text{max}} = 1 \text{ yr}$ capping the step where the ice is slow. The flux divergence is evaluated on the staggered grid with a slope-limited scheme, which suppresses spurious oscillations at the steep thickness gradients near the margin. At every step the surface is updated as $z_s^n = z_b + H^n$ and $\bar{\mathbf{u}}$ is recomputed by the emulator from the current (H, z_s) , so the flux divergence evolves with the geometry rather than being held at its initial-snapshot value. Conventional flux-divergence methods (??), which we retain as a reference in Sect. , instead evaluate $smb = \partial H / \partial t + \nabla \cdot \mathbf{q}$ once from the calibrated state; because the divergence of a flux built from two independently measured fields is itself dominated by small-scale noise, such estimates require *post hoc* spatial smoothing.

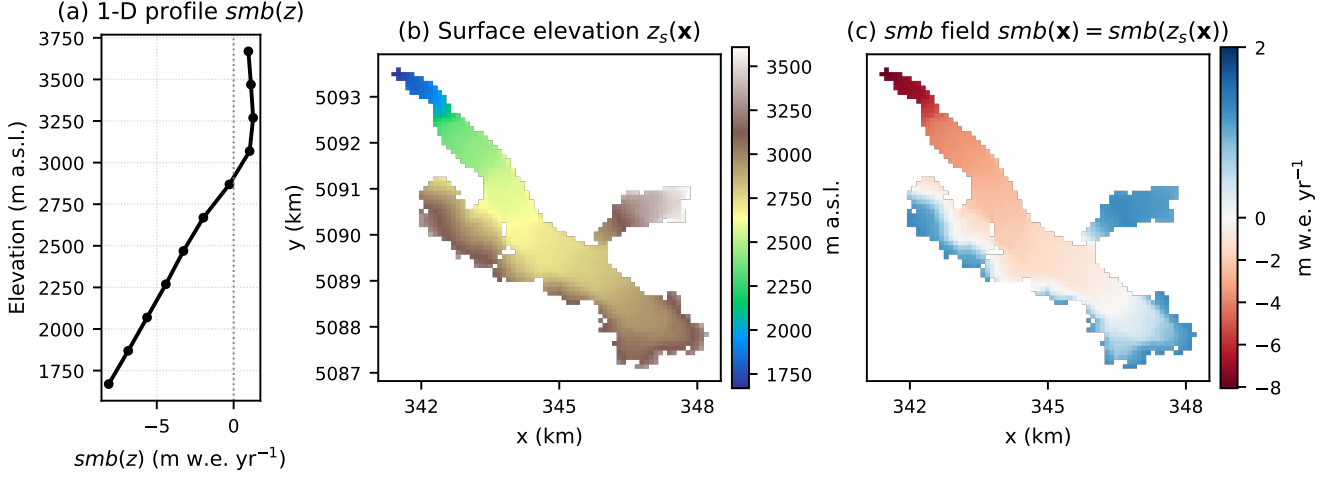


Fig. 2. Mapping the one-dimensional profile to the two-dimensional source term, $smb(\mathbf{x}) = smb(z_s(\mathbf{x}))$, on Argenti re. (a) The inferred elevation profile $smb(z)$; (b) the surface elevation $z_s(\mathbf{x})$; and (c) the resulting 2D smb field, obtained by evaluating the profile at the surface elevation of every grid cell.

SMB parametrisation. Rather than treat the smb as an independent unknown at every grid cell, which would leave the inversion severely underdetermined, we parametrise it as a one-dimensional profile $smb(z)$, a function of surface elevation only. This reflects the first-order elevation control on alpine smb and collapses the unknown from $\mathcal{O}(10^4)$ pixels to $N = \mathcal{O}(10)$ elevation bins, which regularises the otherwise severely ill-posed inversion without site-specific smoothing length scales or density assumptions. The profile is discretised on uniformly spaced elevation bins, 200 m wide for all three glaciers, by the values $\mathbf{m} = (m_1, \dots, m_N)$, and it is mapped to the two-dimensional source term in (1) by evaluating it at the current surface elevation, $smb(\mathbf{x}, t) = smb(z_s(\mathbf{x}, t))$, with piecewise-linear interpolation between bins (Fig. 2). The parametrisation is deliberately minimal: our aim is an smb that explains the observed geometry change, not the resolution of its fine spatial structure.

Inverse problem. We seek the smb profile $\mathbf{m}^*$ that minimises the misfit between the simulated end-of-window thickness $H^{\text{sim}}(\cdot, t_2; \mathbf{m})$ and the target $H^{\text{obs}}(\cdot, t_2)$ from (4):

$$\mathcal{J}(\mathbf{m}) = \sqrt{\frac{1}{|\Omega|} \sum_{\mathbf{x} \in \Omega} (H^{\text{sim}} - H^{\text{obs}})^2} + \mathcal{R}(\mathbf{m}), \quad (7)$$

where the first term is the RMS geometry misfit and $\mathcal{R}(\mathbf{m}) = \lambda \sum_i (m_{i-1} - 2m_i + m_{i+1})^2$ is a Tikhonov smoothness penalty on the squared discrete curvature of the profile. The weight λ carries a direct physical reading: it sets how much the smb may change from one elevation bin to the next, and hence how strongly

the recovered gradient is smoothed. We use $\lambda = 0.1$ throughout; the same value proved optimal for all three glaciers, which suggests it may transfer to alpine glaciers more generally. The minimisation uses L-BFGS (?) with $M = 10$ correction pairs and a Hager–Zhang line search; the low dimensionality of $\mathbf{m}$ makes the implicit Hessian approximation accurate, and convergence is reached in $\mathcal{O}(10^2)$ function evaluations.

The gradient $\nabla_{\mathbf{m}}\mathcal{J}$ is obtained by reverse-mode automatic differentiation (?) through the entire forward integration (1)–(5): every operation in the forward map—the emulator evaluation, the slope-limited divergence, the explicit Euler update and the piecewise-linear elevation interpolation—is a differentiable TensorFlow (?) operation, so the chain rule applies uniformly across the forward graph and the sensitivity of the end-of-window geometry to each bin value is propagated exactly through every time step.

The principal cost is memory. Reverse-mode differentiation must retain intermediate quantities from the forward run in order to traverse it backwards, and a decade-long integration exceeds the memory of a single GPU. We therefore use gradient checkpointing (??): instead of storing the internal quantities of every step, we keep only the glacier state (H, z_s) at the start of each step and recompute that step’s internals when the backward pass reaches it. The saving is substantial because the emulator evaluation—by far the heaviest part of a step—is never stored, only replayed; the price is one additional evaluation of each step per gradient.

RESULTS

We evaluate the method on three Alpine glaciers of contrasting size—Argentière, Great Aletsch and Rhône—each processed through the same glacier-agnostic two-step pipeline: the data assimilation inverts the ice thickness while the basal-motion parameter τ_{ref} is fixed by an ensemble search to the scalar that minimises the surface-velocity misfit, after which the *smb* is inferred. No per-glacier retuning of the regularisation, optimiser or parametrisation is applied, and none of the three uses a thickness survey. Argentière additionally carries the state-of-the-art distributed reconstruction of ?, against which we compare our velocity-only result directly.

Calibrated ice-dynamics states

For every glacier the data assimilation converges within $\mathcal{O}(10^2)$ L-BFGS iterations to a thickness field whose modelled surface velocity matches the observations to a root-mean-square residual of a few m yr^{-1} —2.4 on Argentière, 11 on Great Aletsch and 6.8 on Rhône at the adopted τ_{ref} —one to two orders of magnitude

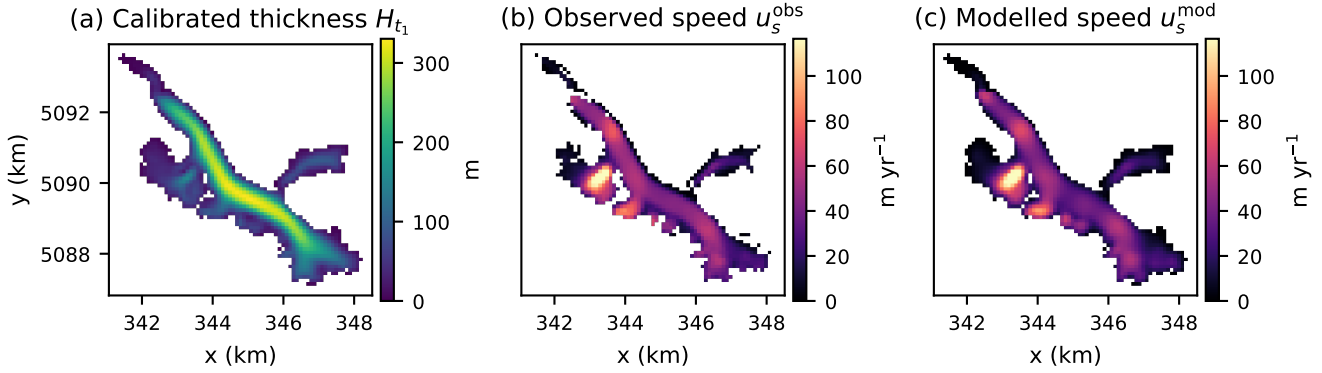


Fig. 3. Calibrated data-assimilation state on Argentière at the start epoch, for the adopted velocity-only run ($\tau_{\text{ref}} = 0.27 \text{ MPa}$). (a) Calibrated ice thickness H_{t_1} ; (b) observed surface speed u_s^{obs} (?); (c) modelled surface speed u_s^{mod} , in close agreement with the observations.

below typical trunk velocities (locally exceeding $100\text{--}150 \text{ m yr}^{-1}$). The calibrated state fixes the bedrock through Eq. (3) and provides the starting geometry H_{t_1} for the transient forward run (Sect.).

Figure 3 shows the calibrated state on Argentière: the recovered ice-thickness field (a), and the close match between the observed (b) and modelled (c) surface speed, whose differences are small and spatially unstructured.

As a final check, we confirm the calibrated state is free of the initialisation shock of Sect. : integrated forward for one year under zero *smb*, it produces no spurious thickness drift, so the glacier is in mechanical equilibrium and any subsequent geometry change is attributable to the prescribed *smb* rather than to residual model imbalance.

Inferred SMB and validation

Holding the calibrated state ($H_{t_1}, \tau_{\text{ref}}, A$) fixed, we run the transient forward simulation over each glacier's window and invert for the elevation-resolved *smb* profile $smb(z)$ that minimises the geometry-misfit cost (Eq. 7). On all three glaciers the retrieved profile increases monotonically across the full altitudinal range, from strongly negative values at the tongue to a positive accumulation regime in the upper basin.

On Argentière (2012–2021) the profile reproduces the observed altitudinal pattern of the GLACIOCLIM stakes with a root-mean-square error of $0.65 \text{ m w.e. yr}^{-1}$ (Fig. 4a, Table 2). This is obtained with the generic velocity-only pipeline—no thickness survey, no per-glacier tuning, no *post hoc* smoothing—yet it is on par with the best physically-based reconstructions of ? on the same stakes, whose SIA and IGM-distributed approaches give 0.67 and $0.85 \text{ m w.e. yr}^{-1}$ and whose flux-gate F2019 estimate gives 0.50 , all

Table 2. Validation of the inferred SMB against the GLACIOCLIM stake network on Argentière Glacier (main trunk + Améthystes tributary) over 2012–2021. The three baselines are the modelling approaches of ?, evaluated against the same stakes; all three use ice-thickness data, whereas our result uses surface velocity only. The GPR-assisted field inversion, which improves further on these, is reported in Appendix ??.

Method	RMSE [m w.e. yr ⁻¹]
F2019 (?)	0.50
SIA (?)	0.67
IGM-distributed (?)	0.85
Ours (velocity only, scalar τ_{ref})	0.65

of which additionally require ice-thickness data. Argentière is also covered by accurate GPR thickness soundings; repeating the experiment with them—inverting τ_{ref} as a spatial field—lowers the error further and surpasses every published reconstruction on this site, as detailed in Appendix ?. Because the inversion variable is one-dimensional, horizontal heterogeneity within an elevation band—notably the avalanche-fed accumulation hotspots reported by ?—is not resolved; this deliberate trade-off, which buys a tuning-free and transferable inversion, is taken up in Sect. .

The same pipeline, applied without modification to Great Aletsch (2009–2017) and Rhône (2007–2016), yields equally smooth, monotonic profiles that reproduce the independent stake averages—two continuous-record locations on Aletsch and five on Rhône (Sect.)—with root-mean-square errors of 0.61 and 0.32 m w.e. yr⁻¹ (Fig. 4b,c). These are on par with the Argentière result despite the absence of any thickness constraint, which we read less as a coincidence than as evidence that the velocity-calibrated dynamical state carries the elevation structure of the SMB; whether the robustness holds across a larger and more varied sample than three large glaciers is an open question we return to in Sect. .

Figure 4 makes concrete the mass-conservation argument of Sect. ?. The dashed curve is the SMB the continuity equation yields when the flux divergence is taken as stationary, exactly as in flux-divergence inversions. On every glacier this estimate is markedly noisier than the inferred profile and diverges in the ablation zone, where the stationary divergence is least representative, raising the stake RMSE to 2.14 (Argentière), 1.59 (Aletsch) and 1.63 m w.e. yr⁻¹ (Rhône)—a factor of 2.6 to 5.1 worse than our transient inversion. Because our forward model integrates the mass-conservation law directly and recomputes the flux divergence from the evolving geometry at every step, the inferred profile requires no spatial smoothing and remains mass-consistent throughout.

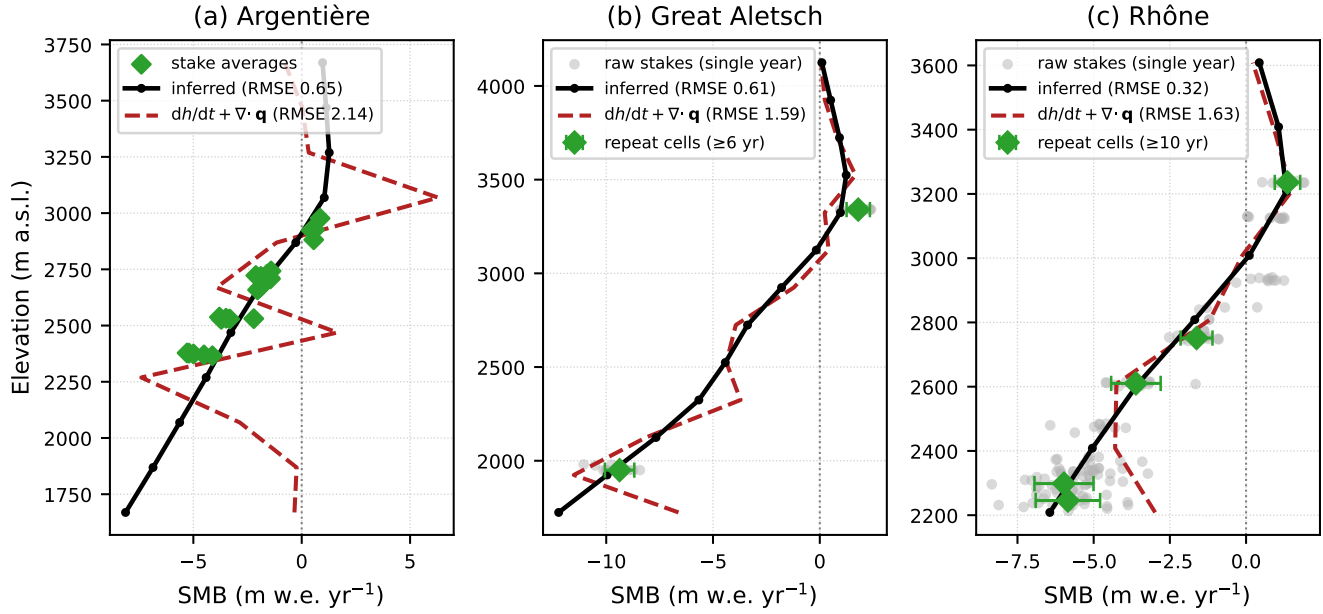


Fig. 4. Inferred SMB profile (black) for (a) Argentière, (b) Great Aletsch and (c) Rhône, compared against in situ stake measurements and against the continuity estimate $dh/dt + \nabla \cdot \mathbf{q}$ (dashed) obtained with the *stationary* data-assimilation flux divergence. Green diamonds are the continuous-record stake averages ($\pm$ interannual standard deviation), the validation targets (faint grey dots on Aletsch and Rhône are the raw single-year readings). The inferred profile tracks the averages across the whole elevation range (RMSE = 0.65, 0.61 and 0.32 m w.e. yr⁻¹), whereas the stationary-flux estimate departs strongly in the ablation zone (2.14, 1.59 and 1.63).

Sensitivity to the basal-motion parameter

The basal-motion parameter τ_{ref} is the dynamical quantity least constrained on a glacier without thickness or stake data, where its value rests on the surface-velocity misfit (*velsurf*) alone. It is also intrinsically hard to separate from the ice thickness: a more slippery bed (smaller τ_{ref}) inferred together with thinner ice reproduces almost the same surface velocity—and hence the same ice flux—as a stiffer bed carrying thicker ice, so many (τ_{ref}, H) pairs fit the observations nearly equally well. For the *smb* inversion this ambiguity is not fatal, because what drives the geometry change is not the pair itself but the flux divergence $\nabla \cdot \mathbf{q}$ it produces. To see how the residual freedom in τ_{ref} propagates into the inferred *smb*, we repeated the *smb* inversion across a wide range of τ_{ref} on all three glaciers (Fig. 5), using the independent stake RMSE as ground truth against *velsurf*, the only selector available operationally.

Two features stand out. First, *velsurf* reliably rejects the strongly biased runs: the profiles that depart most from the stakes are also those with the worst velocity misfit (Fig. 5), so minimising the velocity misfit steers the *smb* inversion away from them—the pipeline discriminates on the misfit alone, not on the magnitude of τ_{ref} . Pooled across the three glaciers, higher velocity misfit goes with higher stake error

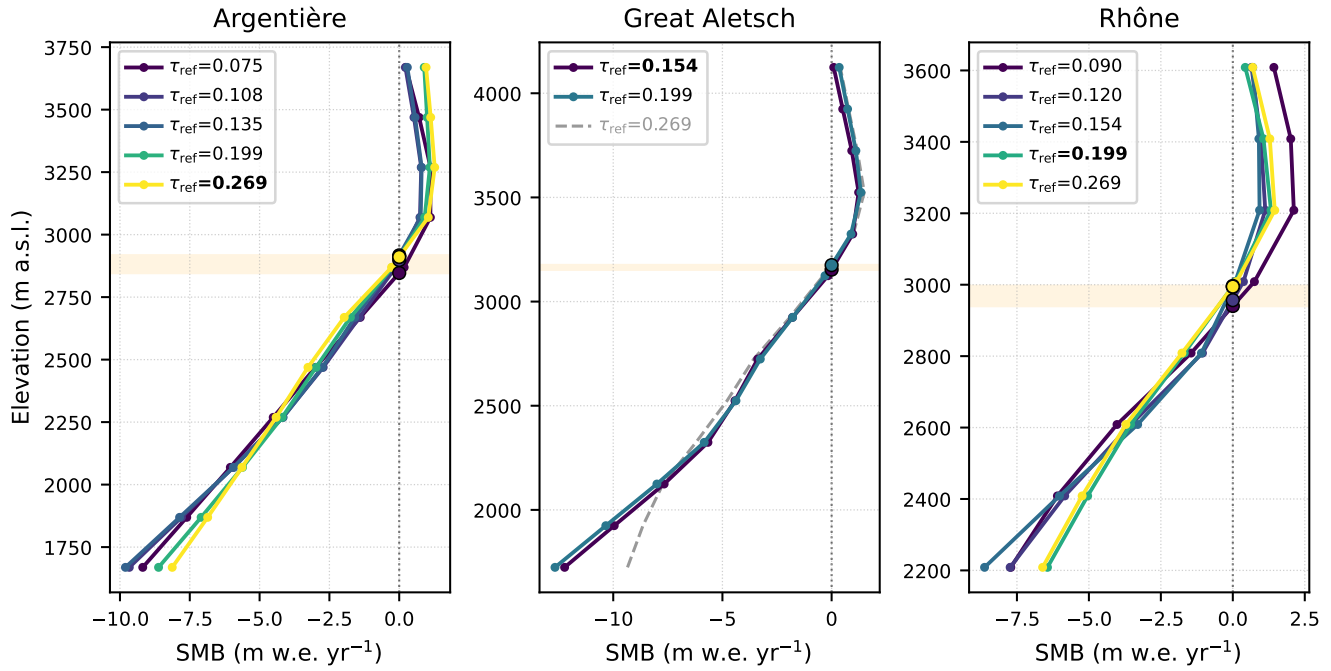


Fig. 5. Inferred SMB profiles across a range of the basal-motion parameter τ_{ref} for (a) Argentière, (b) Great Aletsch and (c) Rhône; colour encodes τ_{ref} and the adopted profile is drawn bold. Within the low-velocity-misfit basin the profiles coincide in the accumulation area and diverge only at the ablation tongue, so the residual uncertainty in τ_{ref} translates into an equilibrium-line-altitude band (shaded) rather than a whole-profile bias; the runs the velocity misfit rejects (dashed) depart most strongly.

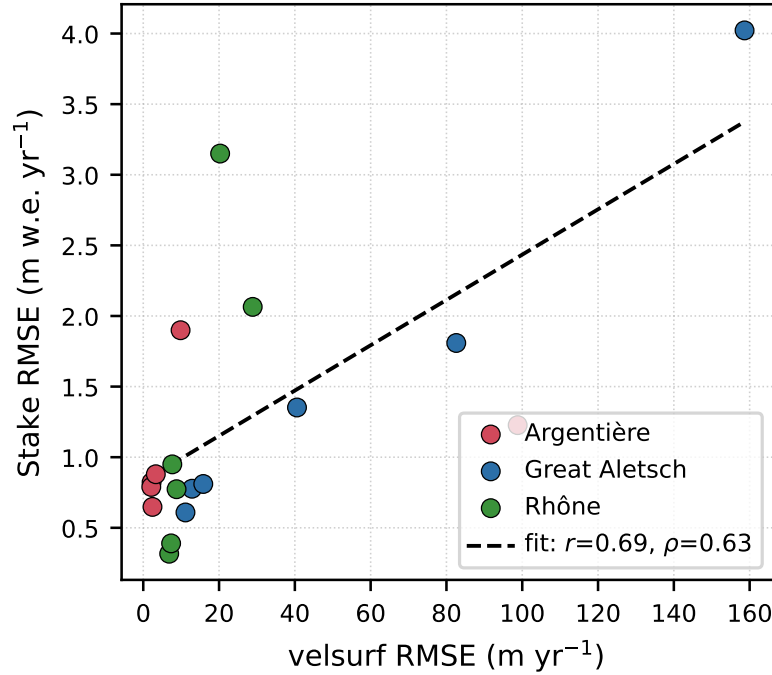


Fig. 6. Data-assimilation surface-velocity misfit (*velsurf* RMSE) against the independent stake RMSE, pooled over the τ_{ref} sweep (six runs per glacier, all three glaciers). Higher velocity misfit is associated with higher stake error (Pearson $r = 0.69$, Spearman $\rho = 0.63$), so the velocity misfit—the only selector available without stakes—carries usable information about the accuracy of the inferred SMB.

(Fig. 6; Pearson $r = 0.69$, $p < 10^{-2}$). Second, once these high-misfit runs are removed, the surviving low-*velsurf* runs no longer map one-to-one onto stake error: because a wide range of (τ_{ref}, H) pairs share nearly the same flux, velocity cannot single out one value of τ_{ref} , and on Rhône the stake RMSE even varies non-monotonically among them (0.32–1.25 m w.e. yr⁻¹). Crucially, this residual freedom is confined to the ablation zone: across the low-misfit runs the accumulation-area profile is essentially unchanged, and only the tongue *smb*—and with it the equilibrium-line altitude—shifts from run to run (Fig. 5).

The practical consequence is that the method does not deliver a single “best” profile but a family of low-*velsurf* solutions that converge to the same accumulation-area gradient and fan out only at the tongue (Fig. 5, Fig. 6). For a stake-free glacier this family is the result: the low-misfit runs together define an *smb* and equilibrium-line uncertainty band—narrow in the accumulation area, widening towards the tongue. How this band behaves at regional scale, and how it might be tightened, we take up in Sect. .

DISCUSSION

The three experiments support a single message: a transient, differentiable inversion recovers the elevation structure of alpine SMB from geometry change and surface velocity alone, without the per-glacier tuning that flux-divergence methods require. On Argentière the generic velocity-only pipeline matches the best physically-based reconstructions of ? on the same stakes, and the optional GPR-assisted field inversion surpasses them (Appendix ??); on the two velocity-only Swiss glaciers the same pipeline, applied without modification, achieves comparable errors. Because the forward model integrates mass conservation directly and recomputes the flux divergence from the evolving geometry, the inferred profile conserves mass exactly and avoids the ablation-zone bias of stationary-flux estimates, which are a factor of two to five worse against the same stakes (Sect.).

The principal limitation is the one-dimensional parametrisation: horizontal variability within an elevation band—such as the avalanche-fed accumulation hotspots resolved by ?—is deliberately not recovered. This is a trade-off of sub-elevation detail for a transferable, tuning-free inversion, and it is not intrinsic to the method: the same differentiable framework admits richer descriptions of the SMB (for example an aspect or debris-cover dependence) wherever the data warrant them, providing a principled route to lift residual non-uniqueness.

Operationally, the sensitivity analysis (Sect.) defines a velocity-only recipe for stake-free glaciers: adopt the τ_{ref} value at the surface-velocity-misfit minimum, and propagate the neighbouring values within the flat misfit basin into an SMB and equilibrium-line uncertainty band. The robustness of the velocity-calibrated state across our three large glaciers is encouraging, but whether it persists across the far larger and more varied sample relevant to regional water-resource applications remains to be tested. Given the rapidly expanding archive of accurate multi-temporal DEMs and near-global velocity products, the tuning-free character of the method makes such a regional-scale, continuous SMB inversion a realistic next step.